

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
MERN Stack

M1:Practical – 2

Roll No.: T025	Name: Afsha Amjad Mujawar
Class: TYIT	Batch: 1
Date of Assignment: 30/06/2026	Date/Time of Submission: 30/06/2026

Aim: Core JavaScript Concepts for MERN

a. Create functions and call them with parameters.

Code:

```
JS practical2a.js > ...
1 function calculateSalary(name,basicSalary) {
2     let bonus = 5000;
3     let totalSalary = basicSalary + bonus;
4     console.log ("Employee Name:",name);
5     console.log ("BasicSalary:",basicSalary);
6     console.log ("Bonus:",bonus);
7     console.log ("Total Salary:",totalSalary);
8 }
9 calculateSalary("Afsha",30000)
```

Output:

```
PS C:\Users\mvlui\Desktop\mern> node practical2a.js
Employee Name: Afsha
BasicSalary: 30000
Bonus: 5000
Total Salary: 35000
PS C:\Users\mvlui\Desktop\mern> 
```

```
1 function generateBill(product,qauntity,price){
2     let total = qauntity * price;
3     let bill= total + (total * 0.09);
4
5
6     console.log ("Product Name:",product);
7     console.log ("Product Quantity:",qauntity);
8     console.log (" price per Iterm :",price);
9     console.log ("Total Bill :",bill);
10    console.log("Afsha");
11 }
12 generateBill("TV",1,40000)
```

Code:

```
PS C:\Users\mvlui\Desktop\mern> node practical2b.js
Product Name: TV
Product Quantity: 1
  price per Item : 40000
Total Bill : 43600
Afsha
```

Output:

b. Work with arrays and objects in JavaScript.

```
1  let fruits = ["mango","apple","banana","cherry"];
2
3  let student = {
4    name:"Afsha",
5    age:20,
6    cty:"Mumbai"
7  };
8
9  console.log("Fruits:",fruits);
10 console.log("Seconf Fruit:",fruits[1]);
11 console.log("Student Name:",student.name);
12 console.log("Student Age:",student.age);
13 console.log("Student City:",student.cty);
14 console.log("Afsha");
```

Code:

```
PS C:\Users\mvlui\Desktop\mern> node practical2b.js
Fruits: [ 'mango', 'apple', 'banana', 'cherry' ]
Seconf Fruit: apple
Student Name: Afsha
Student Age: 20
Student City: Mumbai
Afsha
PS C:\Users\mvlui\Desktop\mern> 
```

Output:

c. Demonstrate use of arrow functions.

```
1  let add = (a,b) => {
2    return a+b;
3  };
4
5  let sub = (a,b) => {
6    return (a-b);
7  };
8  let mul = (a,b) => {
9    return (a*b);
10 };
11 let div = (a,b) => {
12   return (a/b);
13 };
14
15 console.log("Addition:",add(10,20));
16 console.log("Subtraction :",sub(50,20));
17 console.log("Multiplication:",mul(10,20));
18 console.log("Division:",div(40,20));
19 console.log("Afsha");
```

Code:

```

PS C:\Users\mvlui\Desktop\mern> node practical3.js
Addition: 30
Subtraction : 30
Multiplication: 200
Division: 2
Afsha
PS C:\Users\mvlui\Desktop\mern> 

```

Output:

d. Write a program using array methods (map, filter, reduce).

```

1  let number = [10,20,30,40,50];
2
3  let double = number.map(num => num * 2);
4
5  let greater25 = number.filter(num => num > 25);
6
7  let total = number.reduce((sum,num)=> sum + num , 0);
8
9  console.log("Original Array:",number);
10 console.log("Map:",double);
11 console.log("Filter:",greater25);
12 console.log("Reduce:",total);
13 console.log("Afsha");
14

```

Code:

Output:

```

● PS C:\Users\mvlui\Desktop\mern> node prac2d.js
Original Array: [ 10, 20, 30, 40, 50 ]
Map: [ 20, 40, 60, 80, 100 ]
Filter: [ 30, 40, 50 ]
Reduce: 150
Afsha
○ PS C:\Users\mvlui\Desktop\mern> 

```

e. Create a program that manipulates objects and displays output.

Code:

```

JS practical2e.js > ...
1  let student = {
2      roll_No : 25,
3      name: "Afsha",
4      marks: 85
5  };
6
7
8  console.log("Original Object: ");
9  console.log(student);
10
11 console.log("\nAccessing Properties: ");
12 console.log("Name: ",student.name);
13 console.log("Marks: ",student.marks);
14
15 student.marks = 90;
16
17 console.log("\n After Updating Marks: ");
18 console.log(student);
19
20 student.city = "Mumbai";
21 console.log("\n After Adding City: ");
22 console.log(student);
23
24 delete student.roll_No;
25 console.log("\n After Deleting Roll Number: ");
26 console.log(student);
27

```

```

PS C:\Users\mvlui\Desktop\mern> node practical2e.js
Original Object:
{ roll_No: 25, name: 'Afsha', marks: 85 }

Accessing Properties:
Name: Afsha
Marks: 85

After Updating Marks:
{ roll_No: 25, name: 'Afsha', marks: 90 }

After Adding City:
{ roll_No: 25, name: 'Afsha', marks: 90, city: 'Mumbai' }

After Deleting Roll Number:
{ name: 'Afsha', marks: 90, city: 'Mumbai' }
PS C:\Users\mvlui\Desktop\mern>

```

Output:

f. Implement a small JavaScript program combining functions, arrays, and objects.

Code:

```

JS practical2f.js > ...
1  let students = [
2    {
3      roll_No: 100,
4      name: "Rahul",
5      marks: 85
6    },
7    {
8      roll_No: 102,
9      name: "Varun",
10     marks: 85
11   },
12   {
13     roll_No: 103,
14     name: "Anju",
15     marks: 90
16   }
17 ];
18 function display(s)
19 {
20   console.log("Student Details");
21   console.log("-----");
22   for(let stud of s)
23   {
24     console.log(`Roll No: ${stud.roll_No}`);
25     console.log(`Name: ${stud.name}`);
26     console.log(`Marks: ${stud.marks}`);
27     console.log("-----");
28   }
29 }
30 display(students);
31 console.log("Afsha");

```

Output:

```

PS C:\Users\mvlui\Desktop\mern> node practical2f.js
Student Details
-----
Roll No: 100
Name: Rahul
Marks: 85
-----
Roll No: 102
Name: Varun
Marks: 85
-----
Roll No: 103
Name: Anju
Marks: 90
-----
Afsha
PS C:\Users\mvlui\Desktop\mern>

```